IRON (Fe^{2+}) IN BHAVANI RIVER WATER: POTENTIOMETRIC ANALYSIS & POLLUTION ASSESSMENT (CPCB STANDARDS)

Exp. No. :

Date :

AIM/ OBJECTIVE:

To estimate the amount of ferrous ion (Fe^{2+}) present in the whole of the given solution potentiometrically. A standard solution of potassium dichromate of strength ……….N is provided.

PRINCIPLE:

Potentiometric titrations depend on measurement of EMF between reference electrode and an indicator electrode (calomel). When a solution of ferrous ion is titrated with a solution of potassium dichromate, the following redox reaction takes place.

$$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \text{ -------> } 6Fe^{3+} + 2Cr^{3+} + 7H_2O$$

During this titration Fe^{2+} is converted into Fe^{3+} ion and relatively potential of the cell is increases. At the end point, there will be a sharp potential change due to sudden removal of all Fe^{2+} ions. A graph between EMF measured against the volume of $K_2Cr_2O_7$ added is drawn and the end point is noted from the graph as V_1 ml. The cell is set up by connecting this redox electrode with a calomel electrode as shown below.

$$Hg/Hg_2Cl_2(s), KCl//Fe^{2+}, Fe^{3+}/Pt$$

MATERIALS REQUIRED:

Potentiometer, Pt electrode, saturated calomel electrode, glass rod, standard $K_2Cr_2O_7$ solution, ferrous ion (Fe^{2+}) solution.

PROCEDURE:

The given ferrous ion solution is made up to 100 ml in a standard flask. 20 ml of this made up solution is pipetted out into a clean 100 ml beaker. About 10 ml of dil. H_2SO_4 and 10 ml of distilled water are added into it. A platinum electrode is dipped into the solution. This electrode is coupled with a saturated calomel electrode and the cell is introduced into potentiometric circuit. The std. $K_2Cr_2O_7$ solution is taken in the burette and is added in 0.5 ml portions. The emf of the cell is measured after each addition. The addition of $K_2Cr_2O_7$ is continued even after the end point for further 3 ml. The accurate end point is determined by plotting $\Delta E/\Delta V$ Vs Average volume of $K_2Cr_2O_7$ added (graph). From the end point, the strength of ferrous ion solution and its amount can be calculated.

OBSERVATIONS/ INFERENCE:

TITRATION OF $K_2Cr_2O_7$ Vs. FERROUS ION SOLUTION

Volume of Ferrous ion solution = ……..ml

S.No	Volume of $K_2Cr_2O_7$ (ml)	EMF (mV)	ΔE (mV)	ΔV (ml)	$\Delta E/\Delta V$ (mV/ml)	Avg.vol.of $K_2Cr_2O_7$ (ml)
1						
2						
3						
4						
5						
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7						
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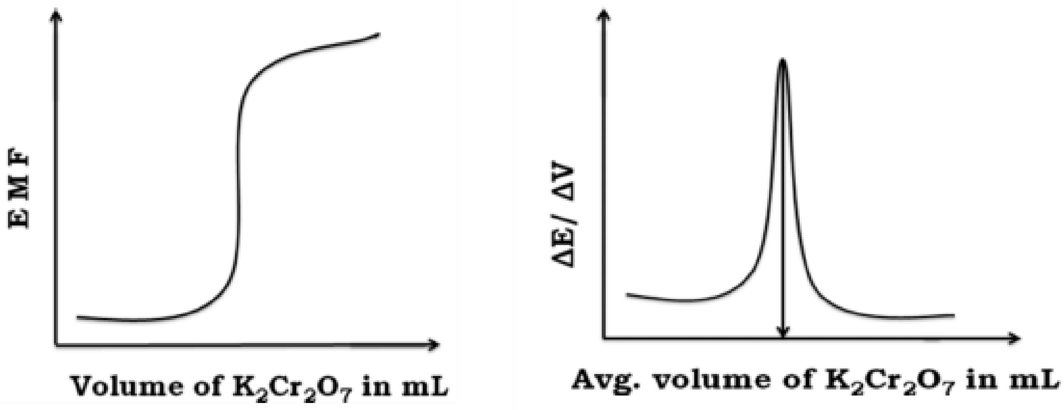
Model graph:



TITRATION OF $K_2Cr_2O_7$ Vs. FERROUS ION SOLUTION

Volume of $K_2Cr_2O_7$ solution	V_1	=	ml (from graph)
Strength of $K_2Cr_2O_7$ solution	N_1	=	N
Volume of Ferrous ion solution	V_2	=	ml
Strength of Ferrous ion solution	N_2	=	?

According to the law of volumetric analysis, $V_1N_1 = V_2N_2$

Strength of Ferrous ion (N_2) $N_2 = V_1N_1 / N_2$

=

Strength of Ferrous ion (N_2) = N

Amount of Ferrous ion present in 1 litre of the given solution = Eq.Wt. x Stength of Ferrous ion

= 55.85 x__________ g/L

=

RESULTS & DISCUSSION:

Amount of Ferrous ion present in 1 litre of the given solution = g/L

MAPPING OF PO & PSO: (For all the COs covered by this experiment)

CO No	PO1	PO2
1	2	1

ASSESSMENT:

Particulars	Max. Marks	Marks awarded
Preparation	10	
Conduct of Experiment	30	
Results & Discussion	30	
Viva - Voce	20	
Report	10	
Total	**100**	
Evaluator's Signature		